

Understanding Freshman Enrollment



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Stat 183
Spring 2017

Outline

1. Introduction

- a. Goal of this Study
- b. Background on the Data

2. Methods

- a. The Model: Logistic Regression with LASSO Penalty
- b. Inference on this model

3. Results & Discussion

- a. High School GPA
- b. Best UC Admitted To
- c. Applicant Location

4. Summary

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Goals of this Study

- Understand what factors into an applicant's decision to attend or not attend UC Riverside.
 - Note: Bernoulli Response
- Build a model that predicts enrollment at UC Riverside.

Background on the Data

- Information on 16301 applicants for the Fall 2014 cycle on 19 variables, some with more than 2 levels.
- Variables:
 - [1] ethnicity, first generation, low income, college applied to, GPA,
 - [6] SAT Score, Academic Index Score, home location, UCD, UCLA,
 - [11] UCSC, UCSC, UCSB, UCI, UCB, UCM,
 - [16] Outreach, Income, Enrolled at UCR
- 25.27% Enrolled, 74.73% Did Not Enroll

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The Model: Logistic Regression with LASSO Penalty

- How it works (Elements of Statistical Learning, Eqn. 4.83):

$$\max_{\beta_0, \beta} \left\{ \sum_{i=1}^N \left[y_i(\beta_0 + \beta^T x_i) - \log(1 + e^{\beta_0 + \beta^T x_i}) \right] - \lambda \sum_{j=1}^p |\beta_j| \right\}$$

Why LASSO?

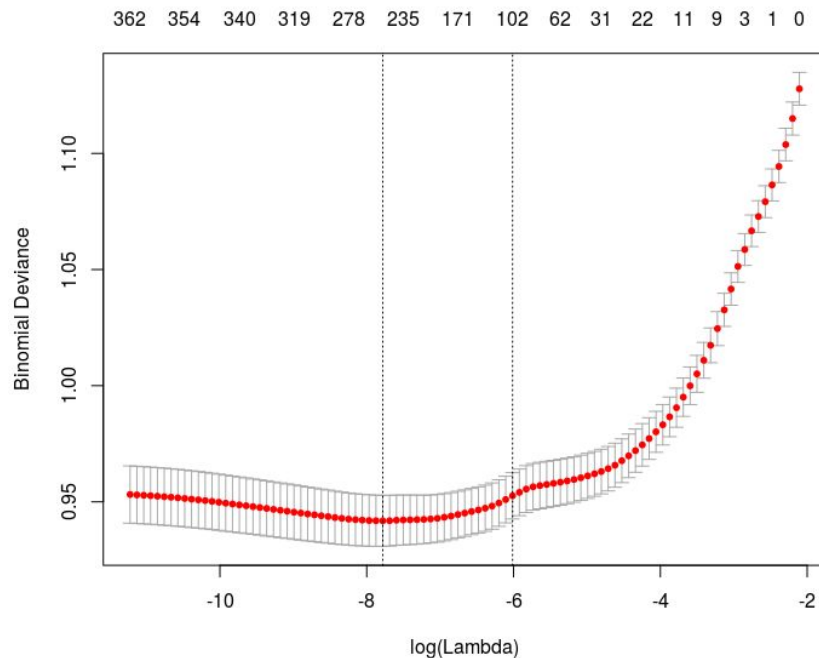
- Personal Preference over stepwise model selection.
 - More stable model parameter estimates with LASSO.
- GLMNET package in R.
 - 2 way interaction terms.

The Model: Logistic Regression with LASSO Penalty

Selecting Regularizing Parameter, λ :

1. 10-fold cross validation.
2. Choose Lambda such that $\text{mean}[\text{deviance}] + \text{se}[\text{deviance}]$ is minimal:

$$\hat{\lambda} = \underset{\lambda}{\operatorname{argmin}} \operatorname{mean}(\{D_{(k)}(y, \hat{\pi}_{\lambda})\}_{k=1}^{10}) + s.e.(\{D_{(k)}(y, \hat{\pi}_{\lambda})\}_{k=1}^{10})$$



Inference on This Model

After selecting the optimal model:

1. For each observation, fit the predicted probability of enrollment at UCR.
2. Examine distribution of the Predicted Probability against a certain explanatory variable.
 - a. Like a “marginal” distribution of predicted probabilities after the effects of the other variables have been taken into account.

NOTE: No direct Inference based on the coefficient of an explanatory variable.

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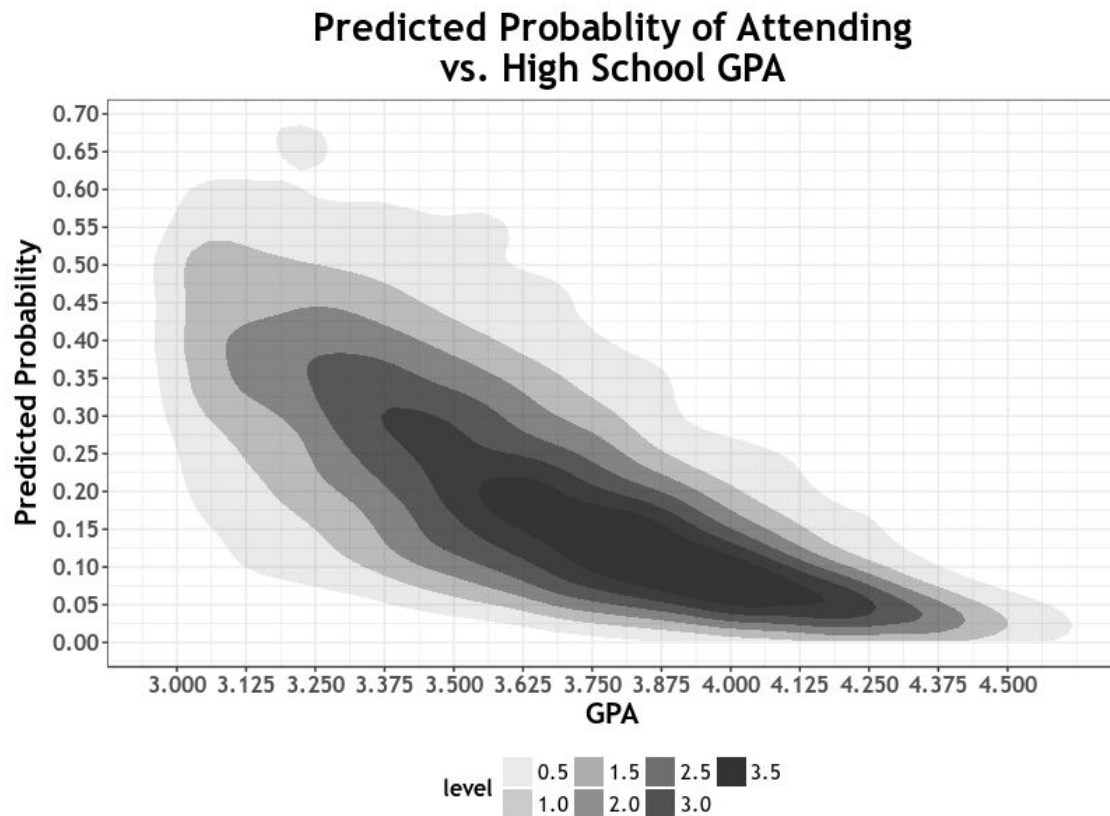
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3. **Results & Discussion**

- a. **High School GPA**
- b. **Best UC Admitted To**
- c. **Applicant Location**

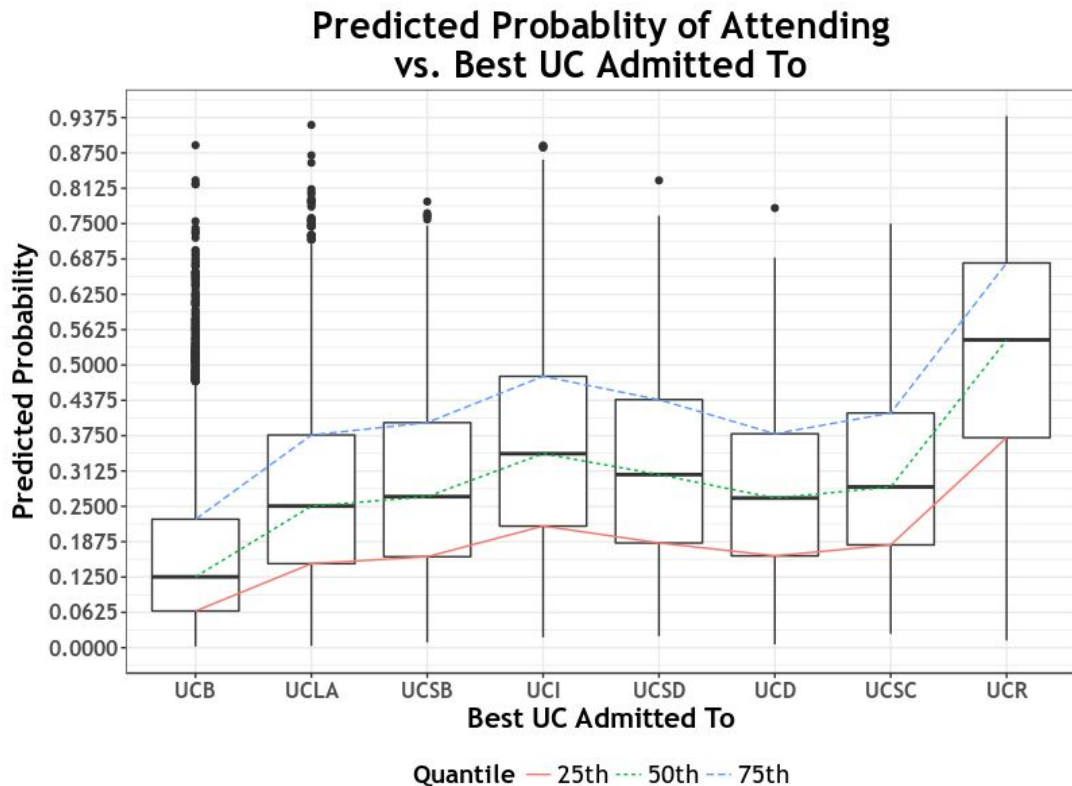
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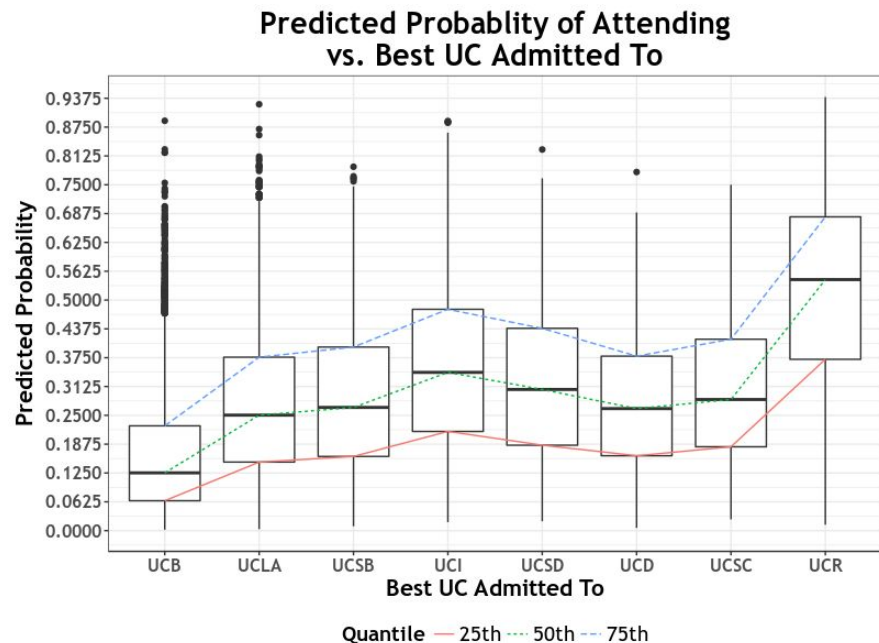
GPA as a Predictor for Enrollment at UCR

- The Higher the GPA, the lower the probability of enrolling at UC Riverside.
- Possible Reason: those students with highest GPAs also have offers from highly sought out universities.
- Similar relationship of predicted probability with SAT score.

Best UC Admitted To as a Predictor for Enrollment



Best UC Admitted To as a Predictor for Enrollment

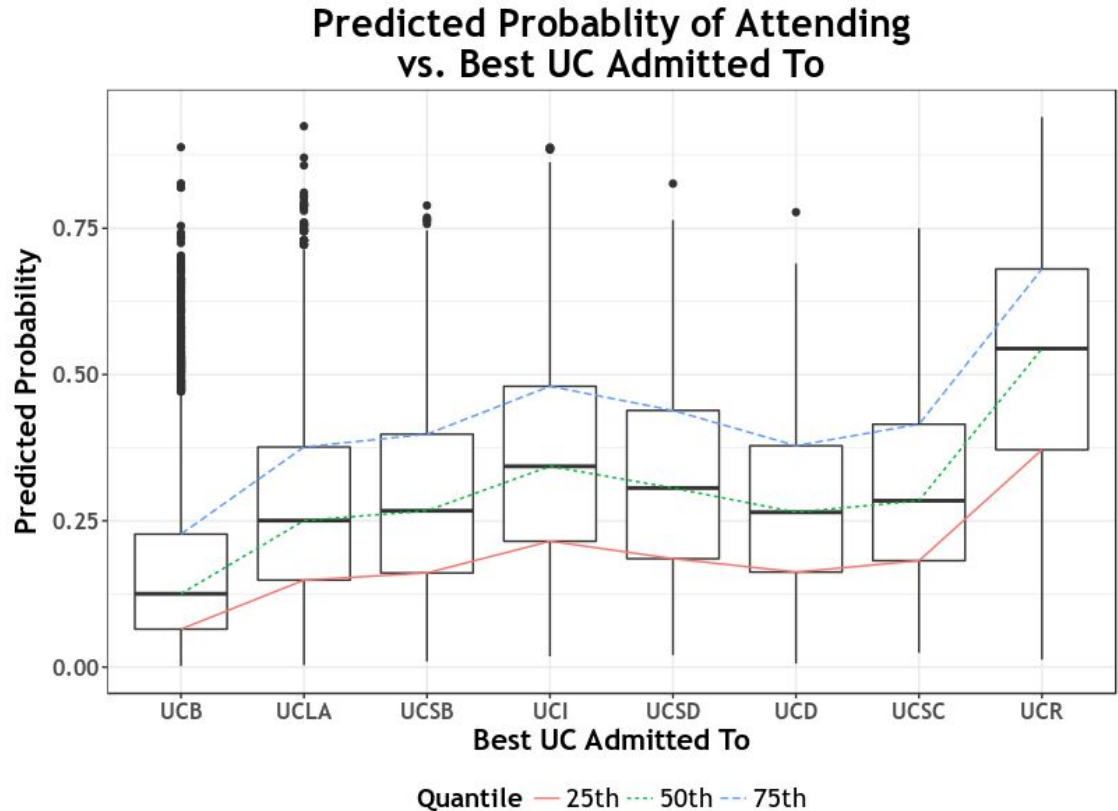


Location	Frequency in Outliers
RIVERSIDE	113
SAN BERNARDINO	60
LOS ANGELES	80
ORANGE	3
SAN DIEGO	0
IMPERIAL	0
CENTRAL CALIFORNIA	0
NORTHERN CALIFORNIA	2
OTHER	7

Table 1: The frequency in Applicant Location among the outliers in the prediction distribution for UCB and UCLA

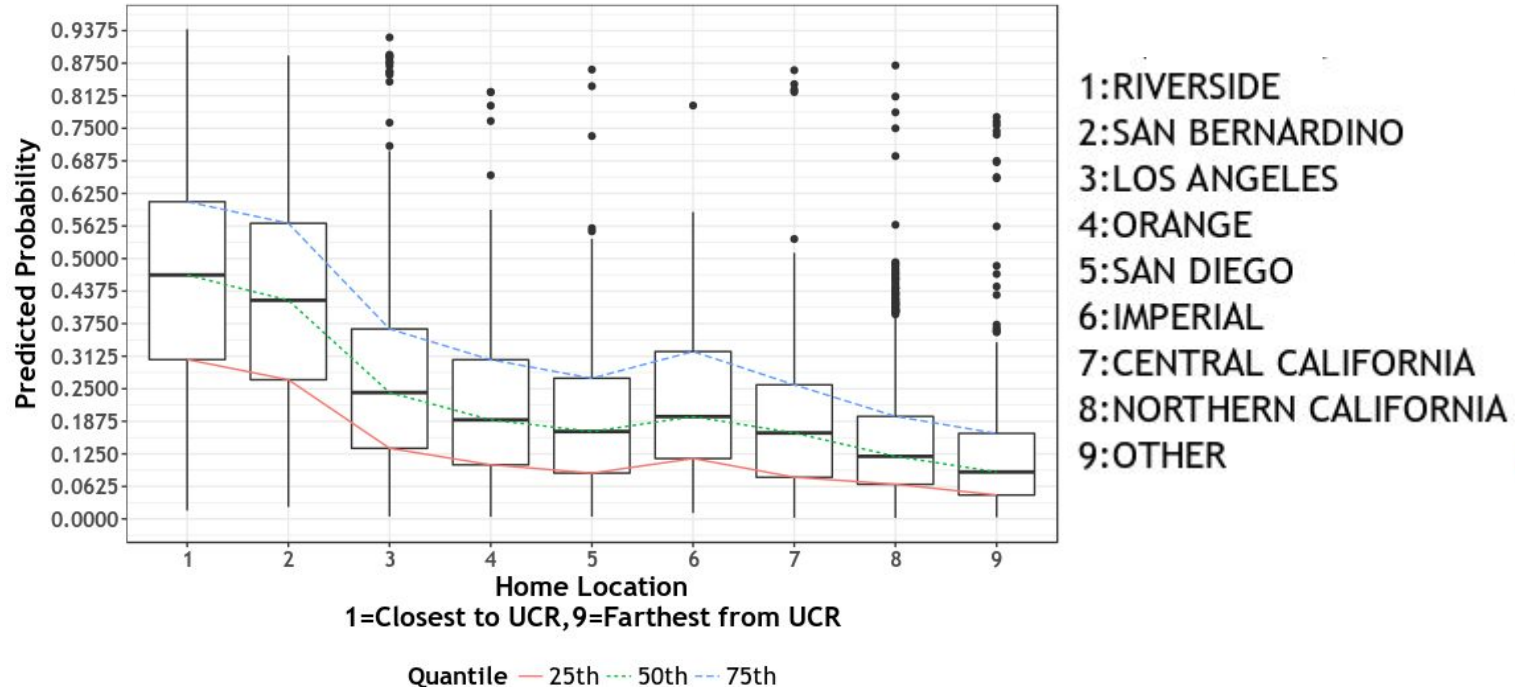
Best UC Admitted To as a Predictor for Enrollment

1. Those who have UC Riverside as the Best UC accepted to are highly likely to attend.
2. Relative to the other schools, those who are accepted to UCI as the best UC have a higher probability of attending UCR.
3. An applicant who is accepted to UC Berkeley is less likely to attend UC Riverside vs. when it is not the Best UC admitted to.
4. However, there are exceptions based on location.



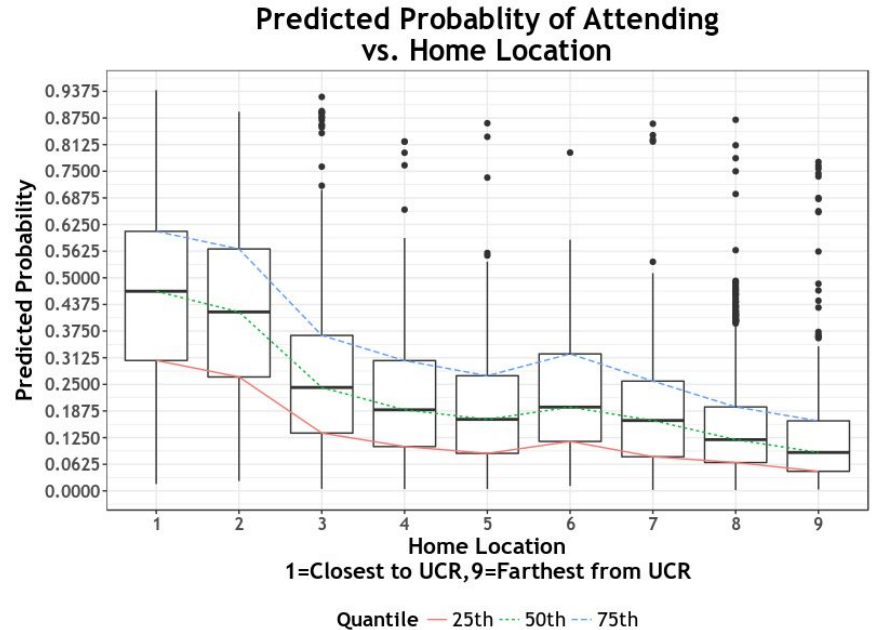
Applicant Location as a Predictor for Enrollment

Predicted Probability of Attending
vs. Home Location



Applicant Location as a Predictor for Enrollment

1. The farther the applicant, the less likely they will attend UC Riverside.
2. However, there are some exceptions.
3. Need more data to understand these exceptions:
 - a. Financial aid?



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Summary: The Main Points

1. Mid to high level GPA students are likely to attend UCR, but those with the highest GPAs are less likely to attend possibly because they have offers from sought after Universities.
2. UCI and UCR seem to compete for the some of the same students.
3. UCB and UCLA admitted applicants are less likely to attend UCR, but there are some outliers who are more likely to attend UCR because they live closer to it.
4. The closer the applicant lives to UCR, the more likely they are to attend.
5. For the furthest applicants (NorCal, Out of State, and International), there are quite a few outliers. This warrants further looking into.

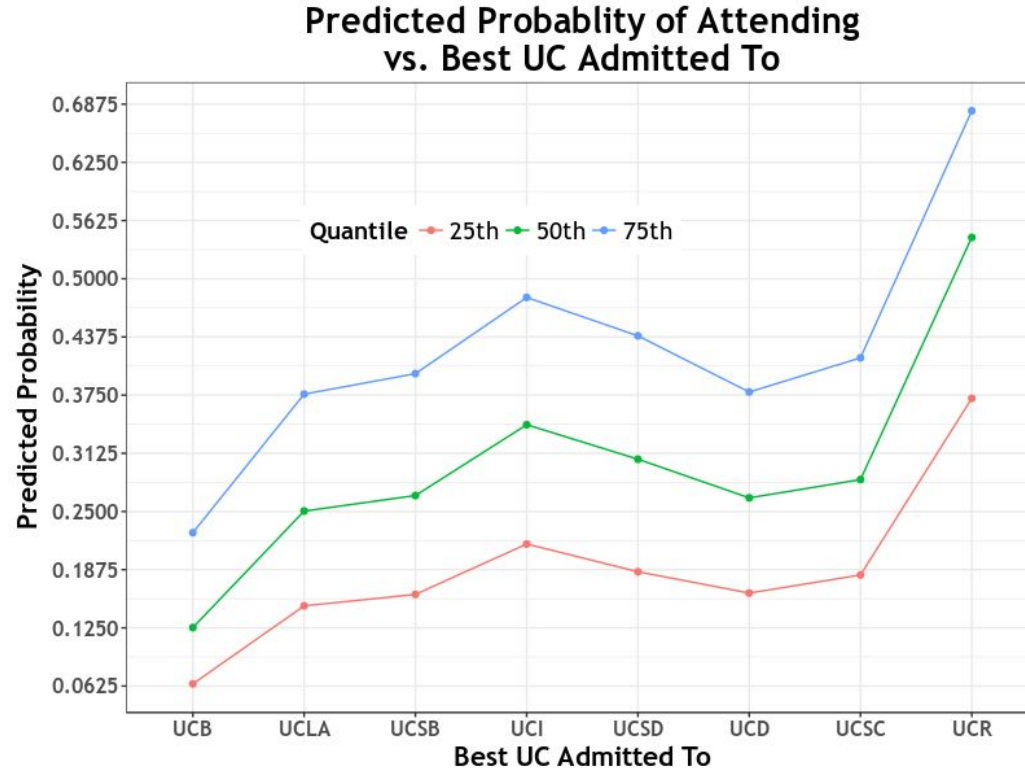
Thank you!

Questions?

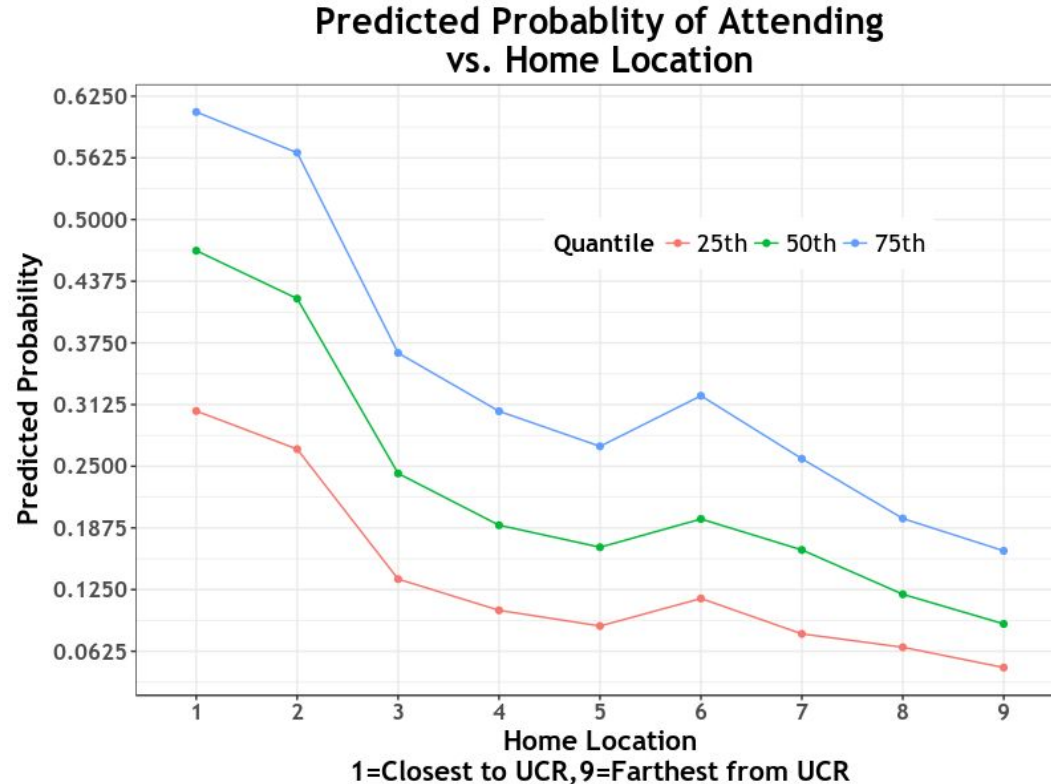
Appendix

Slides that did not make the cut.

Best UC Admitted To as a Predictor for Enrollment

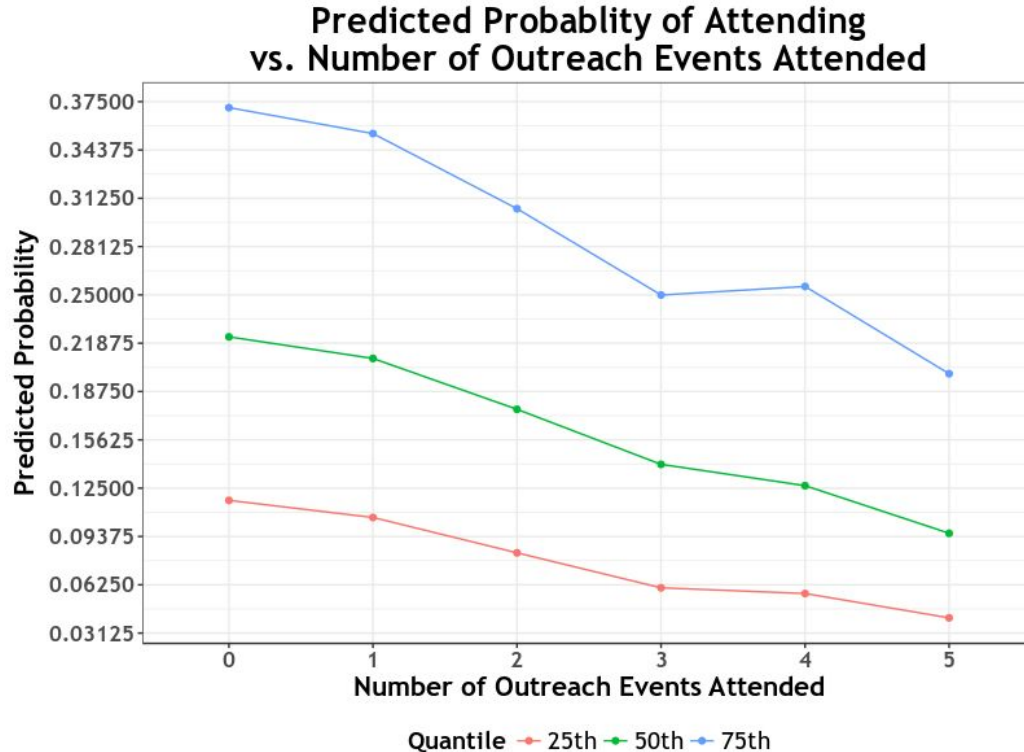


Applicant Location as a Predictor for Enrollment

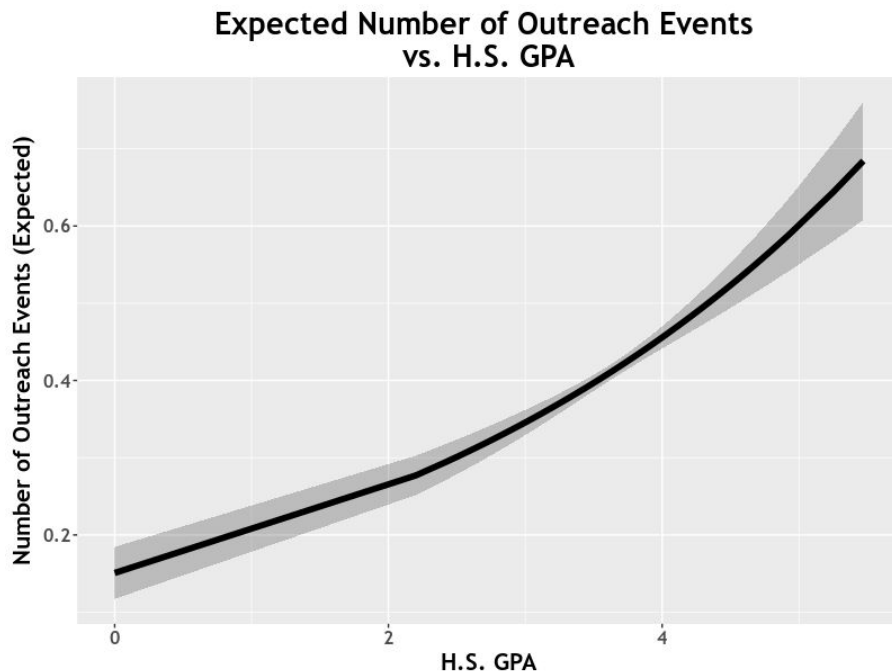


- 1: RIVERSIDE
- 2: SAN BERNARDINO
- 3: LOS ANGELES
- 4: ORANGE
- 5: SAN DIEGO
- 6: IMPERIAL
- 7: CENTRAL CALIFORNIA
- 8: NORTHERN CALIFORNIA
- 9: OTHER

Number of Outreach Events Attended



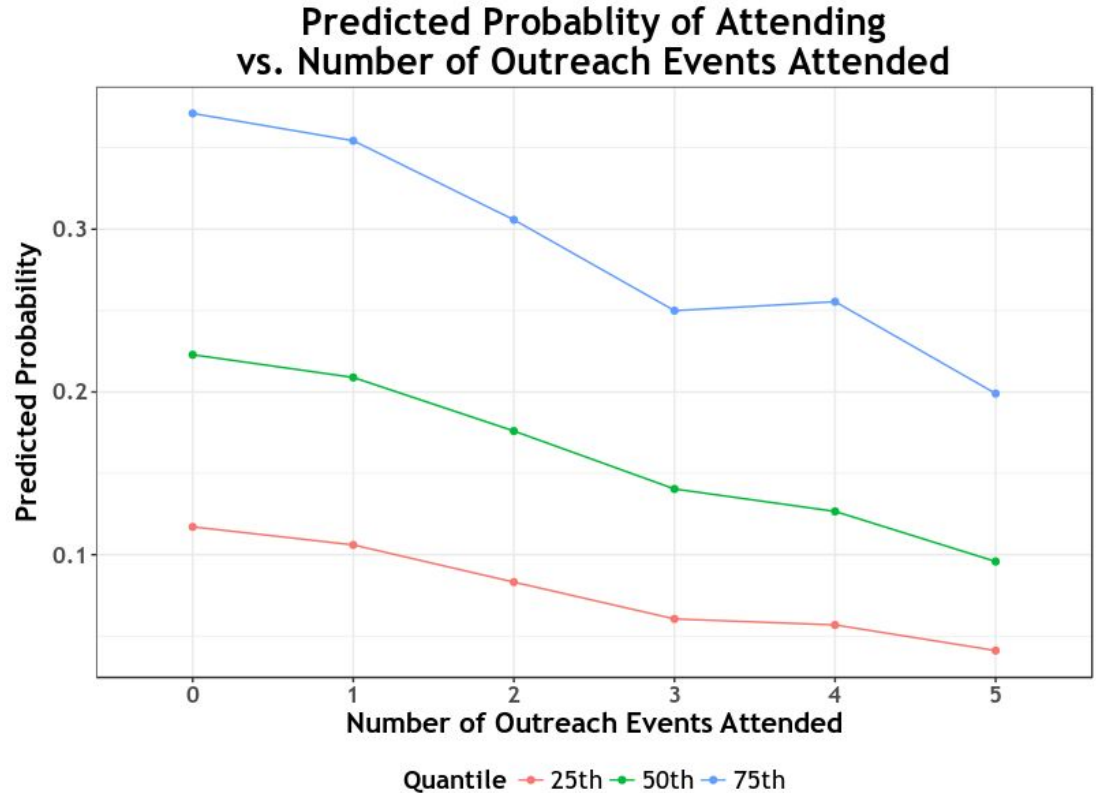
Number of Outreach Events Attended



The higher the GPA, the more outreach events that are expected.

Number of Outreach Events Attended

1. The more outreach events attended, the less likely the applicant will attend.
2. However, there is a confounder: High School GPA.
3. The higher the GPA, the more expected outreach events attended.



Other Findings

1. If an applicant is first generation or low income (or both), they are slightly more likely to attend UC Riverside than if they are not.
2. No Difference between whether applicant applied to CHASS, CNAS, or BCOE.
3. There is evidence of interaction between explanatory variables based on the non-zero coefficient estimates.

Based on the sign of the coefficient estimate in the final model:

- San Bernardino and CNAS = Increased Probability of Attending.
- Imperial Valley and CNAS = Increased Probability of Attending
- Riverside and UCLA = Increased Probability of Attending
- Riverside and UC Davis = Decreased Probability of Attending